Himanshu DigitalTwin

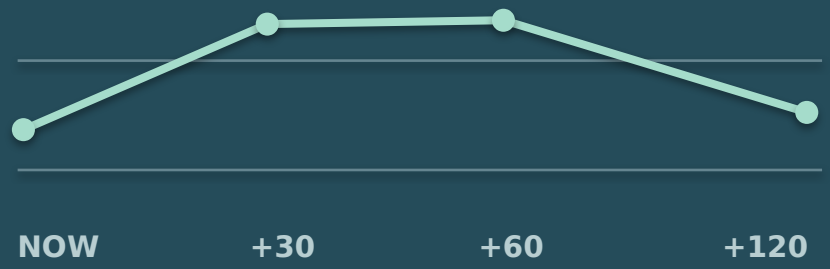
A research prototype for near-term glucose forecasting — with a reason to withhold a prediction.

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Presenter Himanshu Shukla · MITS Gwalior

SYNTHETIC REPLAY / EXAMPLE

144 mg/dL at 12:15



Illustrative numbers. Not a calibrated clinical forecast.

One number describes the present. It does not explain the next two hours.

Patient context and measurement quality change what a forecast can mean.

AT 12:15 PM

144

mg/dL · synthetic observed value

A reading is evidence — not a two-hour prediction.

ILLUSTRATIVE FORECAST FROM THE SAME REPLAY

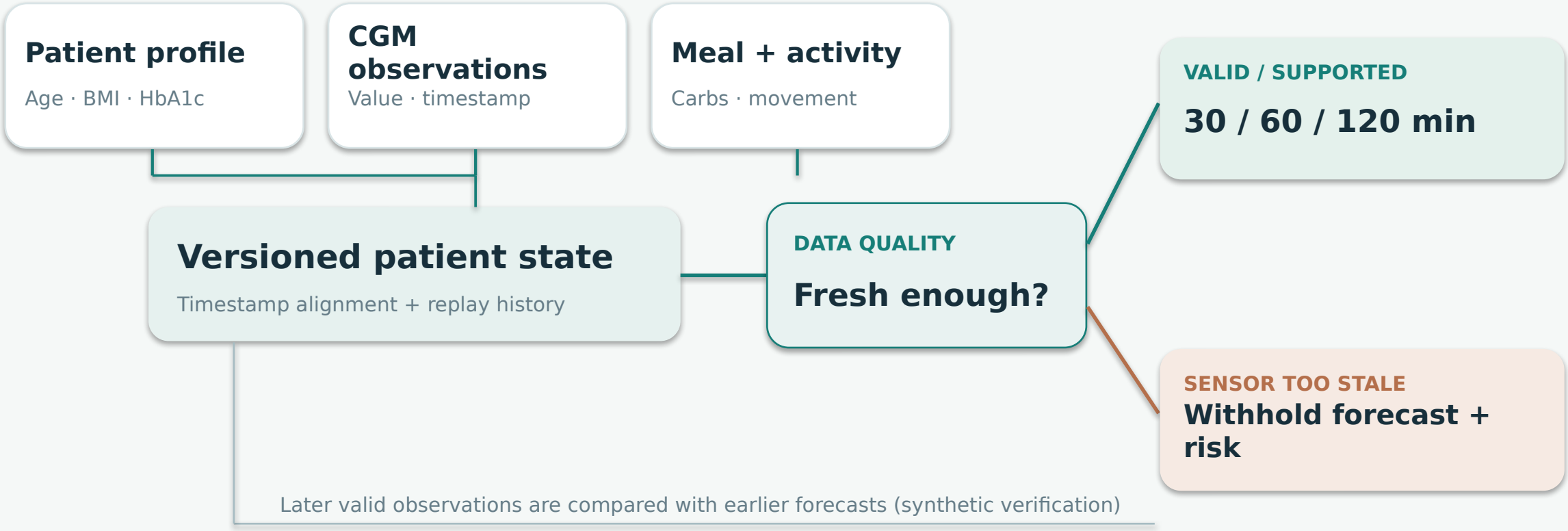
NOW	+30 min	+60 min	+120 min
144	– 155.5	– 156.0	– 145.4

Context: 59.7 g recent carbohydrates · heart rate 82 bpm · time aligned CGM

Question for the prototype: can we show useful context without hiding sensor uncertainty?

Synchronize first. Predict only when supported.

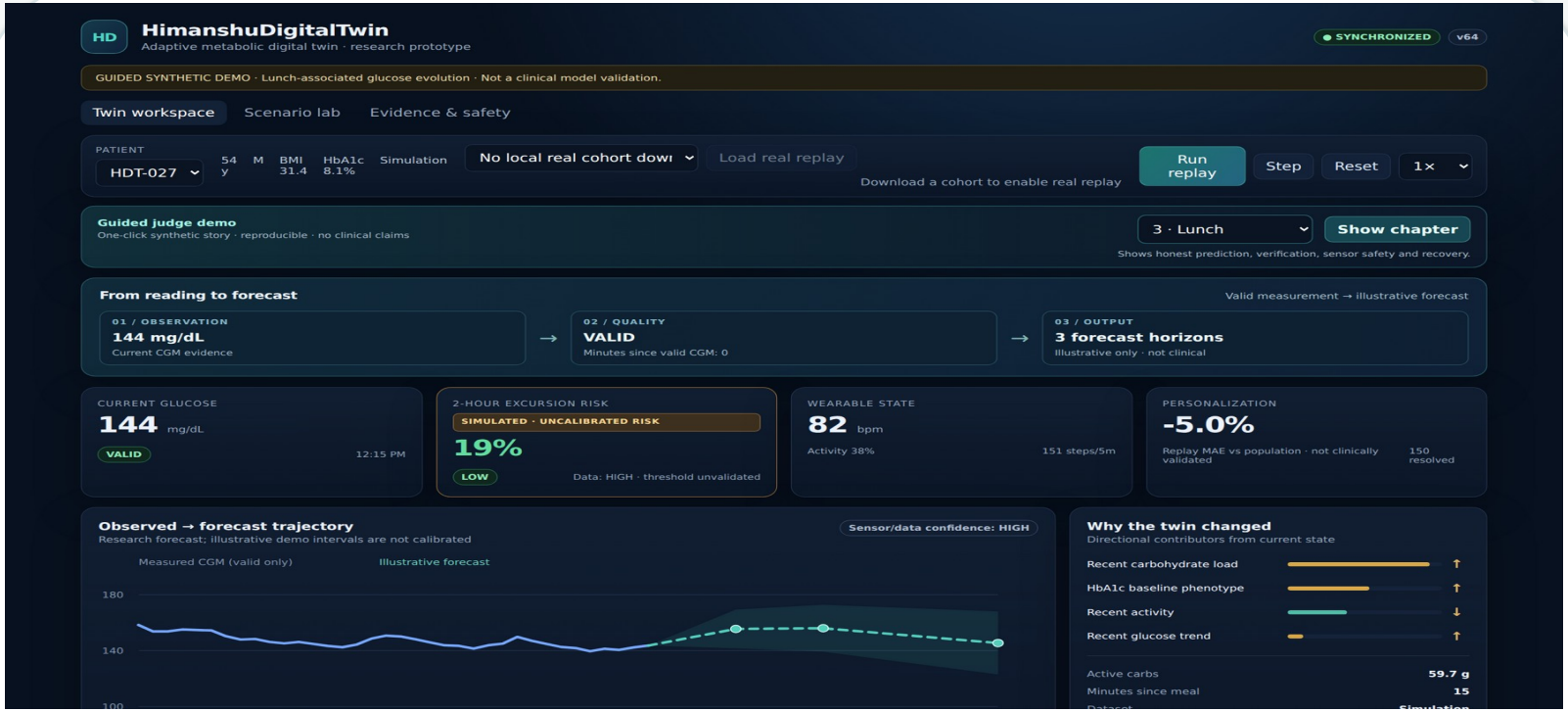
The judge can trace an observation through an explicit quality decision.



Python · FastAPI · HTML/CSS/JS / scikit-learn research pipeline is separate

The running demo — not just an architecture diagram.

Actual FastAPI dashboard, reproducible synthetic lunch checkpoint.



ACTUAL LOCAL DEMO / LUNCH CHECKPOINT

JUDGE PATH

01 · Baseline
Initial reading

02 · Breakfast
Meal context

03 · Lunch
Forecast trajectory

04 · Verification
Forecasts resolved

05 · Sensor gap
Forecast withheld

06 · Recovery
Fresh signal returns

Six deterministic synthetic states. No clinical predictions.

When the signal fails, the interface must say so.

The demo includes a deliberately missing/stale glucose segment.

VALID OBSERVATION

144 mg/dL

Forecast available

Synthetic example at 12:15 PM

STALE SENSOR / 14:25

WITHHELD

No verified glucose · No forecast · No risk

Up to 15 minutes: model may use the last valid reading for a provisional estimate.

Beyond the freshness rule: no estimate is shown. Imputed values are never drawn as measured CGM.

A measured research signal. Not a deployed clinical model.

Independent CGMacros study, patient-grouped holdout; lower MAE is better.

MAE / mg/dL



Gray: persistence baseline Teal: independent study model

EVIDENCE BOUNDARY

45 participants / 14 with T2D

Only 3 T2D participants in the 10-person primary holdout.

The study model is NOT the model running in the synthetic dashboard.

What the judge can verify: deterministic replay, sensor-gap abstention, and research provenance.